SALES PERFORMANCE ANALYSIS AND DASHBOARD

Week 4 Capstone Project

Prepared by – Urja Chauhan

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SALES PERFORMANCE ANALYSIS AND DASHBOARD

Project Objective

The objective of this project is to analyze sales data, identify business trends, build a sales prediction model, and create an interactive Power BI dashboard for business decision-making.

The project demonstrates data cleaning, exploratory data analysis (EDA), predictive analytics, and business intelligence visualization.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn
- Power BI
- Excel

Dataset Description

The dataset contains sales information across different countries, products, and customer segments.

Columns Included:

- Segment
- Country
- Product
- Discount Band
- Units Sold
- Manufacturing Price
- Sale Price

• Gross Sales

• Discounts

• Sales

• COGS

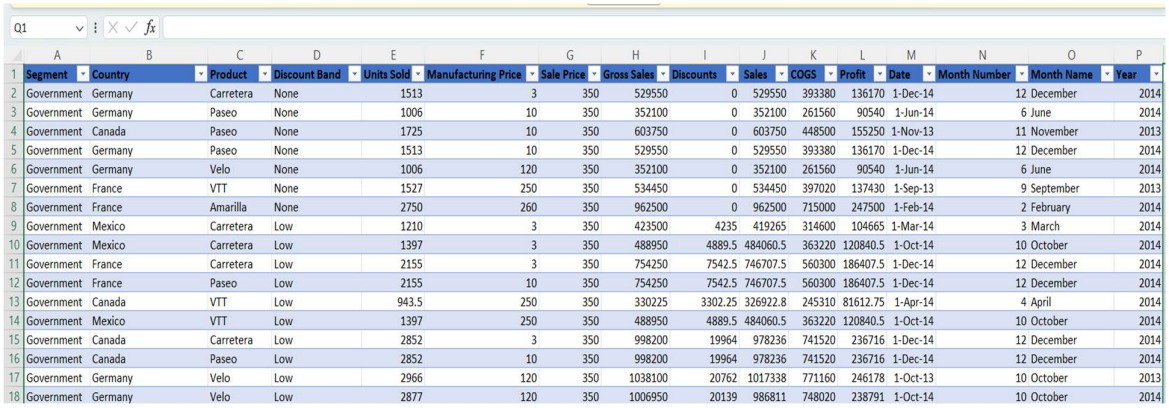
• Profit

• Date

• Month Number

• Month Name

• Year



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Month Name	Year
2	Government	Germany	Carretera	None	1513	3	350	529550	0	529550	393380	136170	1-Dec-14	12	December	2014
3	Government	Germany	Paseo	None	1006	10	350	352100	0	352100	261560	90540	1-Jun-14	6	June	2014
4	Government	Canada	Paseo	None	1725	10	350	603750	0	603750	448500	155250	1-Nov-13	11	November	2013
5	Government	Germany	Paseo	None	1513	10	350	529550	0	529550	393380	136170	1-Dec-14	12	December	2014
6	Government	Germany	Velo	None	1006	120	350	352100	0	352100	261560	90540	1-Jun-14	6	June	2014
7	Government	France	VTT	None	1527	250	350	534450	0	534450	397020	137430	1-Sep-13	9	September	2013
8	Government	France	Amarilla	None	2750	260	350	962500	0	962500	715000	247500	1-Feb-14	2	February	2014
9	Government	Mexico	Carretera	Low	1210	3	350	423500	4235	419265	314600	104665	1-Mar-14	3	March	2014
10	Government	Mexico	Carretera	Low	1397	3	350	488950	4889.5	484060.5	363220	120840.5	1-Oct-14	10	October	2014
11	Government	France	Carretera	Low	2155	3	350	754250	7542.5	746707.5	560300	186407.5	1-Dec-14	12	December	2014
12	Government	France	Paseo	Low	2155	10	350	754250	7542.5	746707.5	560300	186407.5	1-Dec-14	12	December	2014
13	Government	Canada	VTT	Low	943.5	250	350	330225	3302.25	326922.8	245310	81612.75	1-Apr-14	4	April	2014
14	Government	Mexico	VTT	Low	1397	250	350	488950	4889.5	484060.5	363220	120840.5	1-Oct-14	10	October	2014
15	Government	Canada	Carretera	Low	2852	3	350	998200	19964	978236	741520	236716	1-Dec-14	12	December	2014
16	Government	Canada	Paseo	Low	2852	10	350	998200	19964	978236	741520	236716	1-Dec-14	12	December	2014
17	Government	Germany	Velo	Low	2966	120	350	1038100	20762	1017338	771160	246178	1-Oct-13	10	October	2013
18	Government	Germany	Velo	Low	2877	120	350	1006950	20139	986811	748020	238791	1-Oct-14	10	October	2014

Data Cleaning and Preparation

The dataset was imported into Python using Pandas.

Cleaning steps performed:

• Checked data structure

• Verified data types

• Checked missing values

• Generated descriptive statistics

• Prepared data for analysis

```
df.info()
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```
[61]

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 700 entries, 0 to 699
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype         
---  ------                --------------  -----         
 0   Segment               700 non-null    object        
 1   Country               700 non-null    object        
 2   Product               700 non-null    object        
 3   Discount Band         647 non-null    object        
 4   Units Sold            700 non-null    float64       
 5   Manufacturing Price   700 non-null    int64         
 6   Sale Price            700 non-null    int64         
 7   Gross Sales           700 non-null    float64       
 8   Discounts             700 non-null    float64       
 9    Sales                700 non-null    float64       
 10  COGS                  700 non-null    float64       
 11  Profit                700 non-null    float64       
 12  Date                  700 non-null    datetime64[ns]
 13  Month Number          700 non-null    int64         
 14  Month Name            700 non-null    object        
 15  Year                  700 non-null    int64         
dtypes: datetime64[ns](1), float64(6), int64(4), object(5)
memory usage: 87.6+ KB
Number of Rows: 700
Number of Columns: 16
```

```
df.describe()
```
[43]

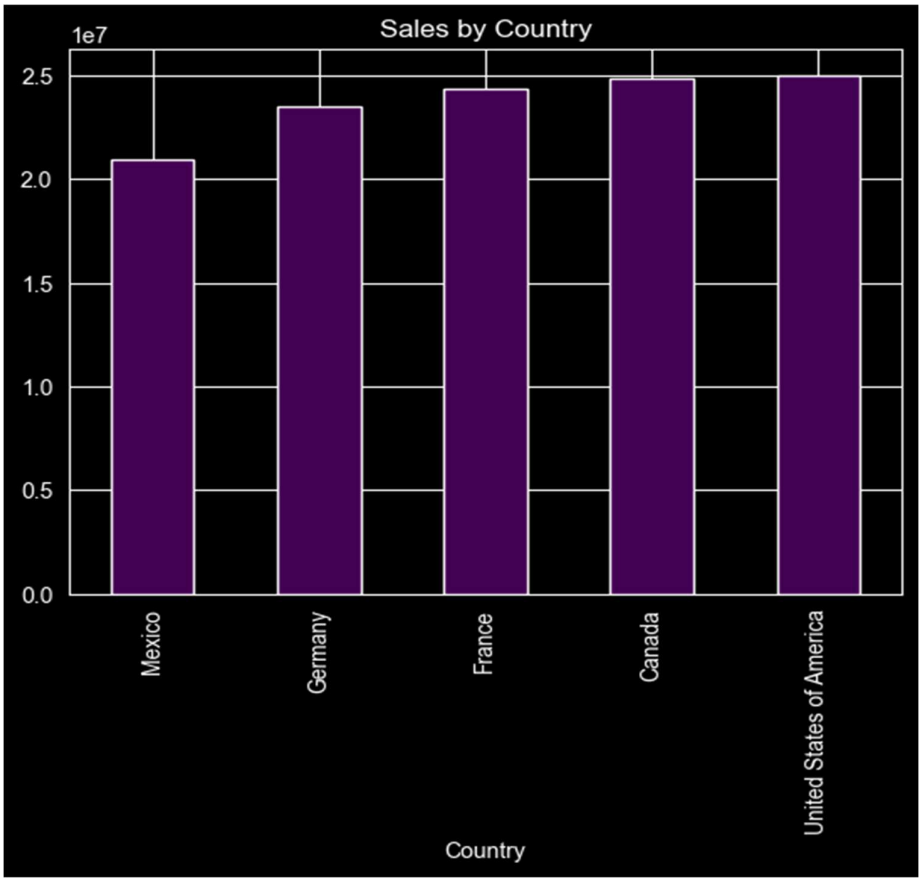
	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Year
count	700.000000	700.000000	700.000000	7.000000e+02	700.000000	7.000000e+02	700.000000	700.000000	700	700.000000	700.000000
mean	1608.294286	96.477143	118.428571	1.827594e+05	13150.354629	1.696091e+05	145475.211429	24133.860371	2014-04-28 21:36:00	7.900000	2013.750000
min	200.000000	3.000000	7.000000	1.799000e+03	0.000000	1.655080e+03	918.000000	-40617.500000	2013-09-01 00:00:00	1.000000	2013.000000
25%	905.000000	5.000000	12.000000	1.739175e+04	800.320000	1.592800e+04	7490.000000	2805.960000	2013-12-24 06:00:00	5.750000	2013.750000
50%	1542.500000	10.000000	20.000000	3.798000e+04	2585.250000	3.554020e+04	22506.250000	9242.200000	2014-05-16 12:00:00	9.000000	2014.000000
75%	2229.125000	250.000000	300.000000	2.790250e+05	15956.343750	2.610775e+05	245607.500000	22662.000000	2014-09-08 12:00:00	10.250000	2014.000000
max	4492.500000	260.000000	350.000000	1.207500e+06	149677.500000	1.159200e+06	950625.000000	262200.000000	2014-12-01 00:00:00	12.000000	2014.000000
std	867.427859	108.602612	136.775515	2.542623e+05	22962.928775	2.367263e+05	203865.506118	42760.626563	NaN	3.377321	0.433322

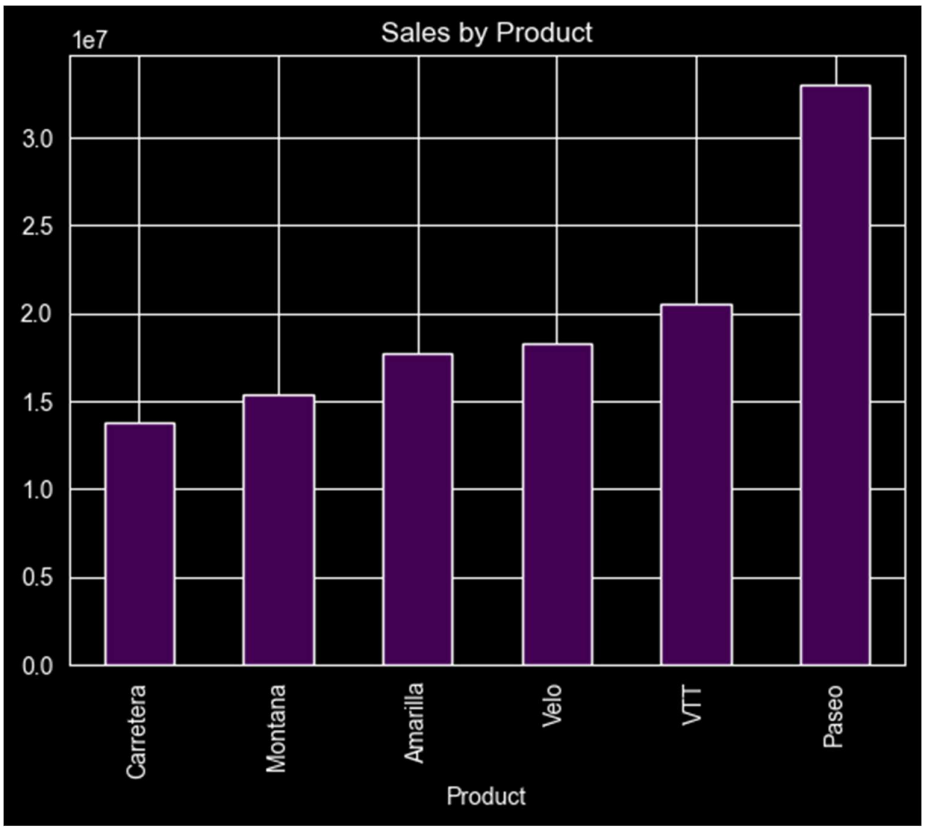
Exploratory Data Analysis (EDA)

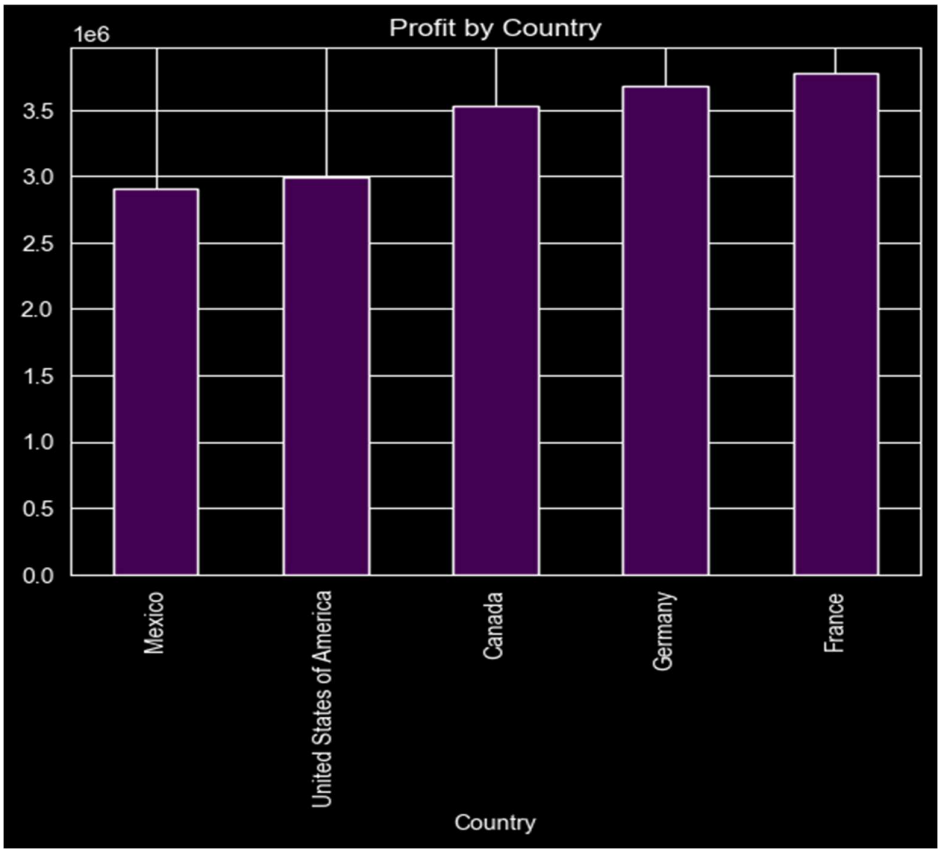
EDA was performed to understand business performance and identify trends.

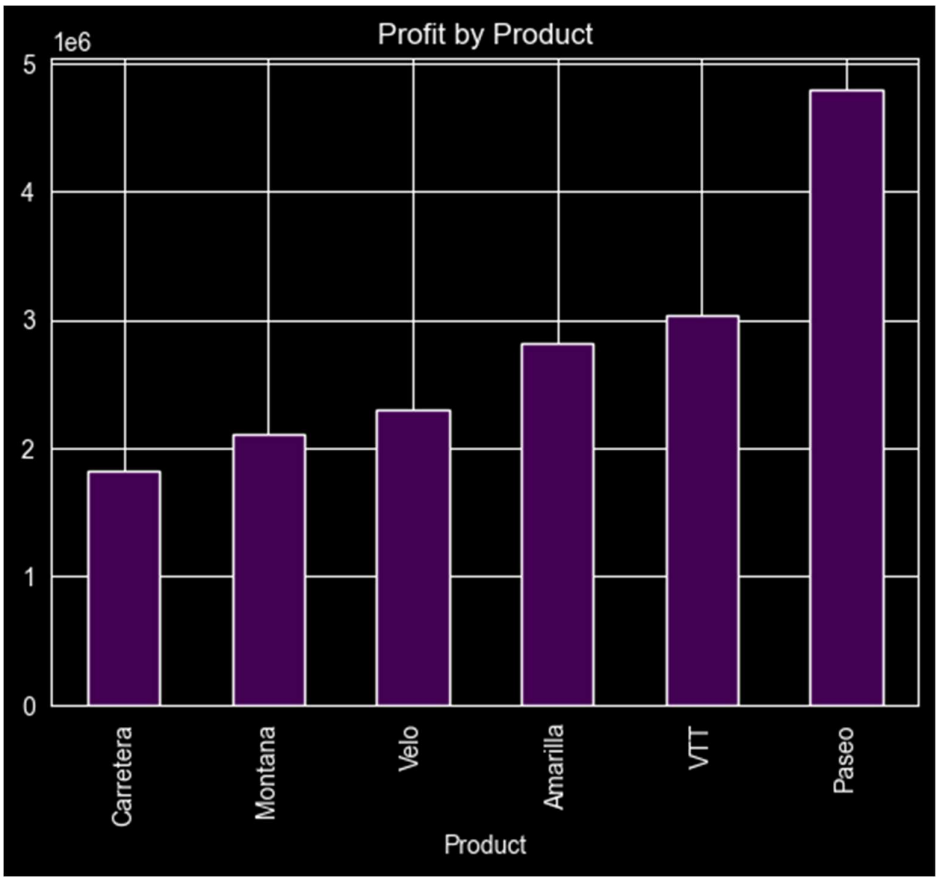
Analyses Performed:

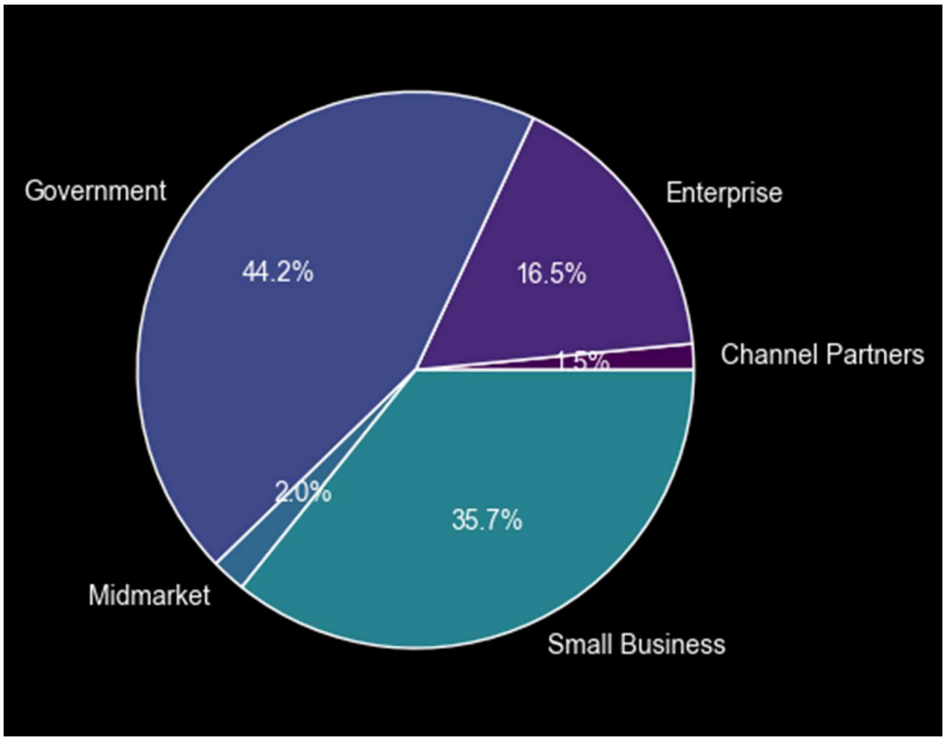
- Country-wise sales analysis
- Product-wise sales analysis
- Segment-wise sales analysis
- Monthly sales analysis
- Yearly sales analysis
- Profit analysis
- Correlation heat map



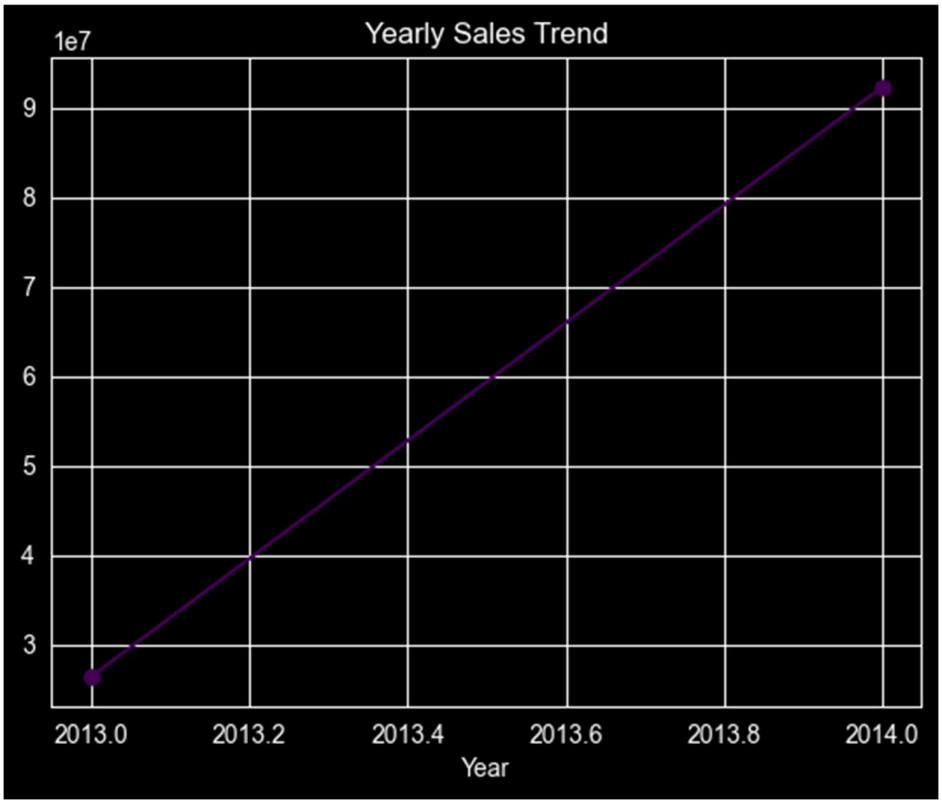

Sales by Product
1e7
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Carretera
Montana
Amarilla
Velo
VTT
Paseo
Product

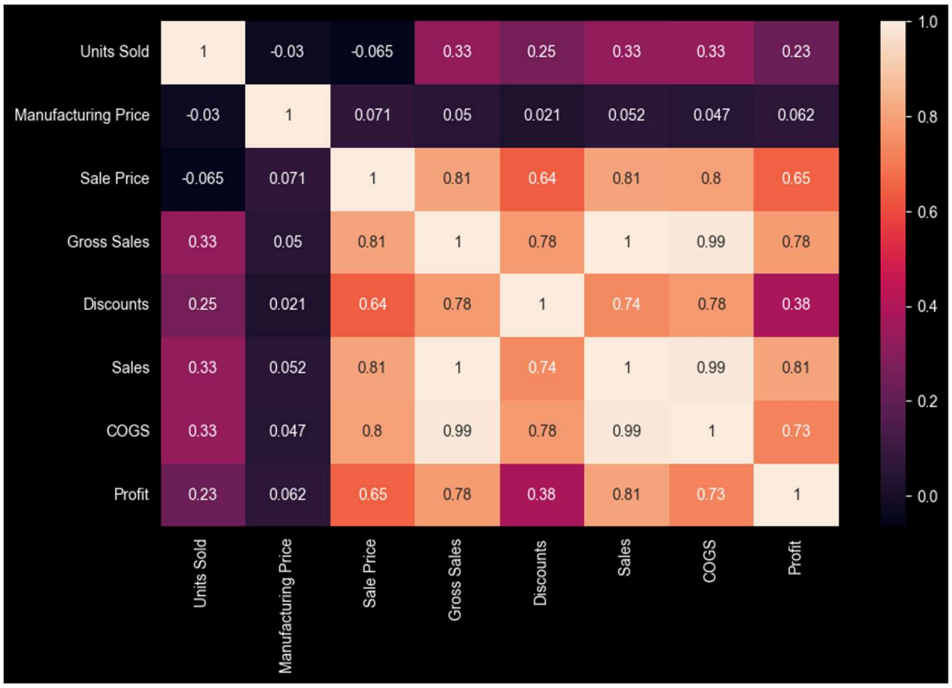

Profit by Country
1e6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Mexico
United States of America
Canada
Germany
France
Country


Profit by Product
1e6
5
4
3
2
1
0
Carretera
Montana
Velo
Amarilla
VTT
Paseo
Product


Government
44.2%
Enterprise
16.5%
Channel Partners
1.5%
2.0%
35.7%
Midmarket
Small Business


Monthly Sales Trend
1e7
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
2
4
6
8
10
12
Month Number


Yearly Sales Trend
1e7
9
8
7
6
5
4
3
2013.0
2013.2
2013.4
2013.6
2013.8
2014.0
Year



Sales Prediction Model

A Linear Regression model was developed to predict sales.

Process:

- Feature Selection
- Data Splitting
- Model Training
- Model Testing
- Prediction Generation

Evaluation Metrics:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

```
X = df[["Units Sold",
        "Manufacturing Price",
        "Sale Price",
        "Discounts",
        "COGS",
        "Profit"]]
```
[52]

```
Y = df[" Sales"]
```
[53]

```
from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.2, random_state = 42)
```
[54]

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, Y_train)
```
[55]

▾ LinearRegression

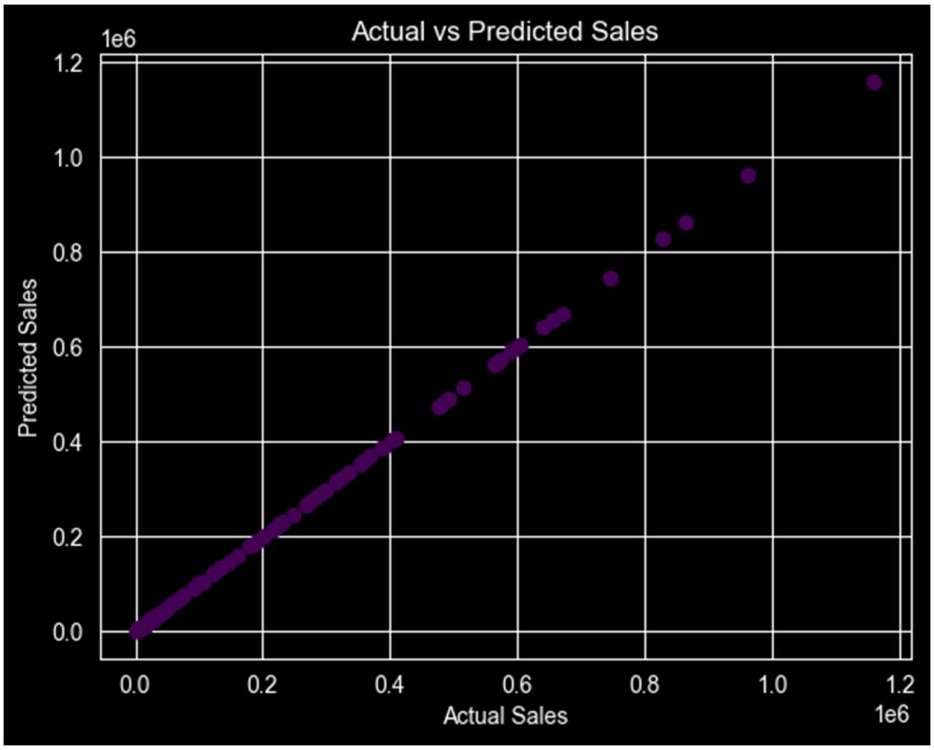
▸ Parameters

▸ Fitted attributes

```
Y_pred = model.predict(X_test)
```
[56]

```
from sklearn.metrics import r2_score
print("R2 Score:", r2_score(Y_test, Y_pred))
```
[57]

```
R2 Score: 1.0
```

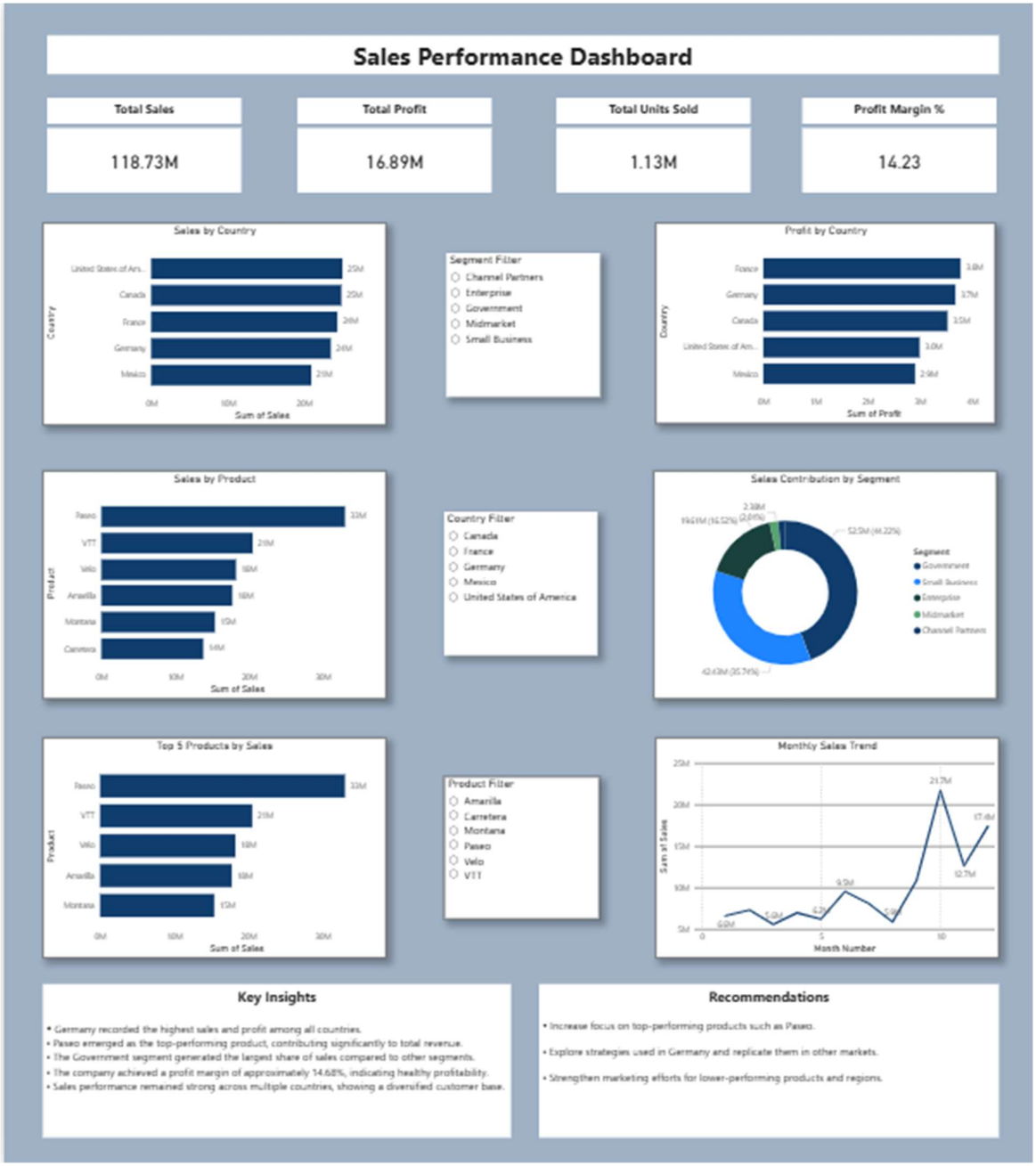


Power BI Dashboard

An interactive dashboard was developed using Power BI to monitor business performance.

Dashboard Components:

- Total Sales KPI
- Total Profit KPI
- Total Units Sold KPI
- Profit Margin KPI
- Sales by Country
- Profit by Country
- Sales by Product
- Segment Contribution
- Top 5 Products
- Interactive Filters



Key Insights

• Germany recorded the highest sales and profit among all countries.

• Paseo emerged as the best-performing product.

• Government segment contributed the largest share of total sales.

• The company maintained a profit margin of approximately 14%.

• Sales performance remained strong across multiple markets.

Recommendations

• Increase focus on top-performing products such as Paseo.

• Expand successful strategies used in Germany to other regions.

• Improve marketing efforts for low-performing products.

• Continue monitoring KPIs using interactive dashboards.

• Use predictive models to support future sales planning.

Conclusion

This project successfully demonstrated the complete data analytics workflow, including:

• Data Cleaning

• Exploratory Data Analysis

• Sales Prediction using Machine Learning

• Dashboard Development using Power BI

The generated insights can help stakeholders make informed decisions and improve business performance.
